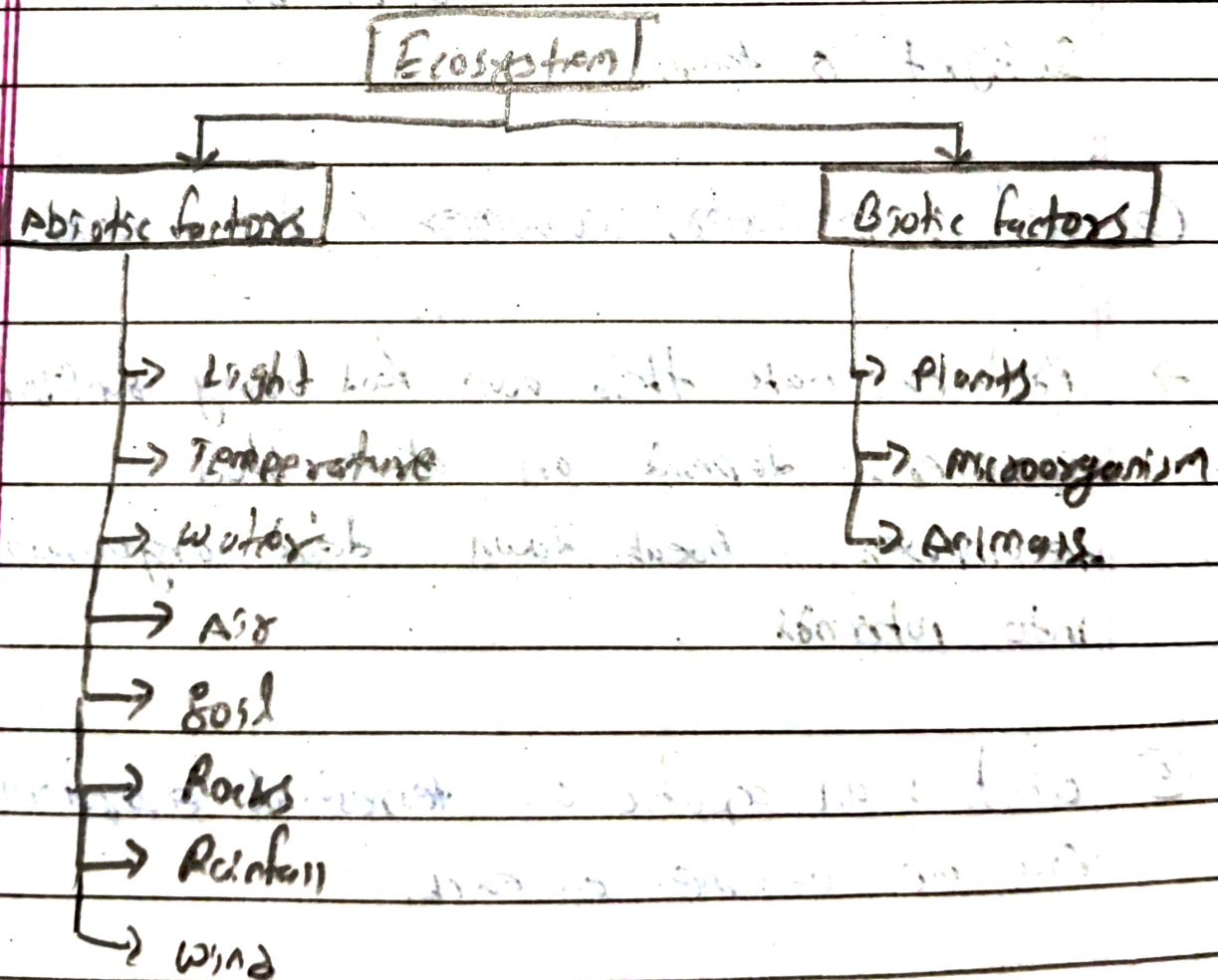


Assignment: 1

- ① Explain the structure of an ecosystem with a neat diagram.

→ An ecosystem has biotic (living) and abiotic (non-living) components. Biotic includes producers, consumers, and decomposers. Abiotic includes air, water, soil and sunlight.



② Define biotic and abiotic components with examples.

→ Biotic components: are living organisms like plant, animal and microorganisms.

Abiotic components: are non-living factors that are, water, soil, sunlight & temp.

③ Define producer, consumer & decomposers.

→ Producer: make their own food using sunlight
consumer: depend on other food

decomposer: break down dead organisms into nutrients

④ What is an aquatic and terrestrial ecosystem?
Give me example on each.

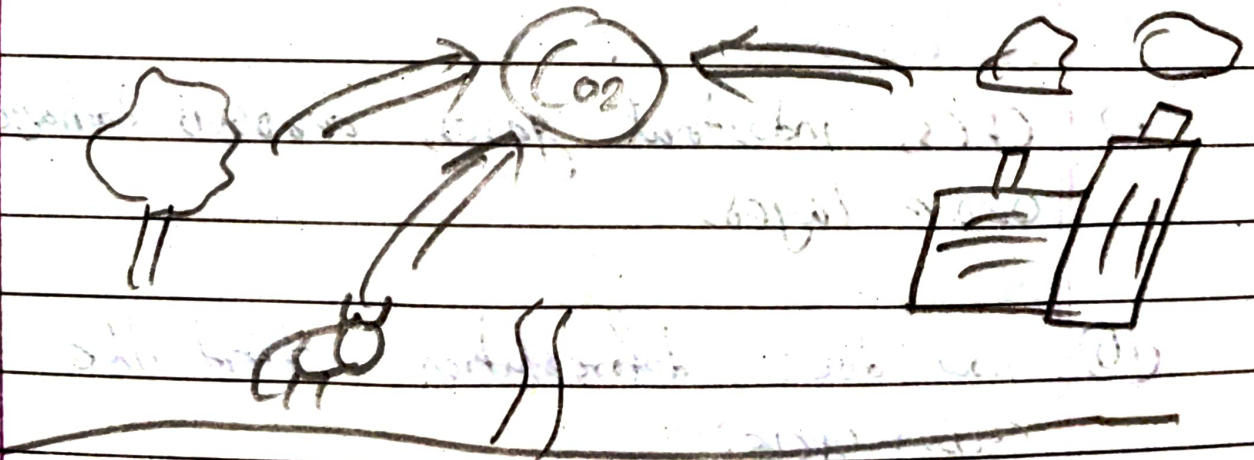
→ An aquatic ecosystem exists in water bodies like pond, river or oceans. Terrestrial ecosystem exists on land like forest or desert.

⑤ Differentiate between food chain and food web in examples.

- A food chain is a linear sequence of energy flow.
A food web is a network of interconnected food chains, providing multiple feeding paths and stability.

⑥ Explain the carbon cycle with a diagram.

- The carbon cycle is the movement of carbon in nature. Plants absorb CO_2 during photosynthesis. Animals release CO_2 through respiration. Decomposers return carbon from dead matter to the atmosphere. Burning fuels also adds CO_2 .



By: carbon cycle

⑦ What is global warming? List its main causes.

→ Rise in Earth's temperature due to greenhouse gases: causes; fossil fuels, deforestation, pollution.

⑧ Explain the process of the greenhouse effect.

→ Greenhouse gases trap heat from sunlight, warming Earth.

⑨ Discuss the causes of ozone depletion.

→ CFCs, industrial gases, aerosols damage ozone layer.

⑩ How does deforestation affect the carbon cycle?

→ Reduces CO_2 absorption - increases atmospheric CO_2 - causes warming.

Ques:

① How does ozone depletion affect human health and the environment?

→ Ozone depletion allows harmful UV rays to reach Earth, it cause skin cancer, eye damage, weak immunity in humans.

② Explain the structure and functioning of an ecosystem in detail.

→ The structure includes biotic & abiotic components. Functioning involves energy flow, nutrient cycling, and interactions between organisms.

③ Explain the importance of food web in maintaining ecological balance.

→ Food web provide multiple feeding relations, maintain population balance, and increase ecosystem stability.

④ Explain global warming in detail, including causes, effects & prevention

→ Global warming is caused by greenhouse gases from fossil fuels and deforestation. Effects include rising temp, melting glaciers, and extreme weather. Prevention involves using renewable energy, reducing emissions, and planting trees.

⑤ Discuss ozone depletion; causes, effects, & prevention measures.

→ Causes include CFCs and industrial pollutants. Effects include increased UV radiation causing health and environmental damage. Prevention includes reducing CFC use, using eco-friendly products & spreading awareness.

Assignment: 2

① Compare and contrast classical smog (London type) & Photochemical smog (Los Angeles type).

→ Classical smog forms in cold, humid conditions due to coal burning (SO_2 , smoke), while photochemical smog forms in warm, sunny conditions due to vehicle emission (NO_x , hydrocarbons).

② Distinguish between Criteria Pollutants and Hazardous Air pollutants (HAPs). Which organization defines these categories?

→ Criteria Pollutants are common pollutants with set air quality standards (e.g. CO , SO_2). HAPs are toxic pollutants causing serious health effects (e.g. benzene). These categories are defined by the US Environmental Protection Agency (EPA).

③ Explain the biochemical mechanism by which CO affects human health.

→ Carbon monoxide binds with hemoglobin to form carboxyhemoglobin, reducing oxygen transport in blood, this leads to oxygen deficiency, causing dizziness, unconsciousness or even death.

④ Scientifically distinguish between "sound" and "noise".

→ Sound is any audible vibration with organized waves, while noise is unwanted or unpleasant sound that causes disturbance.

⑤ Explain why the decibel (dB) scale is logarithmic rather than linear.

→ The decibel scale is logarithmic because sound intensity varies over a wide range. It compresses large variations into manageable values and reflects how humans perceive sound.

⑥ Discuss ~~and~~ the Psychological and Physiological impacts of noise pollution on human health, including its effects on sleep patterns, cardiovascular health (blood pressure), and cognitive performance in children.

→ Noise Pollution disturbs sleep, increases stress and blood pressure, and may lead to heart problems. In children, it reduces concentration, memory and learning ability.

Assignment: 3

① Explain the significance of BOD and COD parameters in determining water quality

→ BOD measures oxygen needed by microbes to decompose organic matter, while COD measures oxygen needed to oxidize all pollutants. Higher value indicate higher pollution.

② Describe the complexometric titration method using EDTA to determine the total hardness of a water sample. Why is a buffer solution (pH approx 10) essential for this experiment?

→ EDTA binds with Ca^{2+} and Mg^{2+} ions, changing the indicator color from wine red to blue at endpoints. Buffer (pH ~ 10) maintains proper conditions for accurate reaction & sharp color change.

③ outline the three stages of municipal wastewater treatment).

→ Primary: removal of solids, secondary: biological decomposition; tertiary: removal of nutrients and disinfection

④ Discuss how the excessive use of nitrogenous fertilizers leads to soil acidification

→ Nitrogen fertilizers releases H^+ ions during nitrification, lowering soil pH and reducing fertility

⑤ Explain how persistent organic pollutants (POPs) such as certain pesticides, move through the Soil-Plant-Human food chain

→ POPs enter soil, are absorbed by plants then pass to animals and humans through food accumulating at each level (biomagnification)

Assignment: 4

SAGAR

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① Define Renewable and Non-renewable resources.
Provide three examples for each category.

→ Renewable resources are naturally replenished in a short time. ex: Solar energy, wind energy, hydropower

Non-renewable resources are limited and take millions of years to form

examples:

coal, petroleum, natural gas.

② compare the energy density of fossil fuel (like coal or oil) to renewable sources (like solar or wind energy).

→ Fossil fuels have high energy density, producing more energy per unit.
Renewable sources have lower energy density and require larger area or systems to generate the same energy.

③ Explain the diff b/w photovoltaic system and concentrated solar power.

→ PV systems convert sunlight directly into electricity using solar cells.
CSP uses mirrors to concentrate sunlight to produce heat, which generates electricity via turbines.

④ Why is geothermal energy considered "location-specific"?

→ It depends on underground heat sources, which are available only in certain regions like volcanic or tectonic areas.

⑤ Explain why certain resources like soil or fisheries are sometimes considered "conditionally renewable".

→ These resources can renew naturally, but only if used sustainably. Overuse or mismanagement can make them non-renewable.

Assignment 5

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1) Integrated solid waste management (ISWM).

→ A comprehensive approach to manage waste using reduction, reuse, recycling, treatment, & safe disposal to minimize environmental impact.

2) Hauled vs stationary container system.

→ Hauled - container system (HCS): container is transported to disposal site

• Stationary - container system (SCS): container stays, waste is collected and taken away.

3) Describe the physical and chemical characteristics of municipal solid waste (MSW).

→ Physical: Density, moisture content, particle size, calorific value

• Chemical: organic matter, carbon content, nitrogen, toxic elements

4) Differentiate between E-waste and Biomedical waste. Suggest an AI-based tracking system to prevent the illegal dumping of hazardous industrial waste.

→ E-waste: Discarded electronic items like computers and phones

• Biomedical waste: waste from hospitals such as syringes and human tissues

AI-based system: use IoT sensors, GPS tracking, and AI algorithms to monitor waste movement, and prevent illegal dumping.

⑤ As an AIML student, define "Green AI". How does the carbon footprint of training a large-scale Deep learning model relate to Environmental Management Systems?

→ Green AI refers to developing AI models that consume less energy and reduce environmental impact.

The carbon footprint of large models is high due to intensive computation, so Environmental management systems help monitor, reduce, and optimize energy usage.

⑥ Discuss the salient features of the Solid waste management Rules (2016) in India.

- Segregation of waste at source (wet, dry, hazardous)
- Door to door collection
- Ban on open dumping and burning.

1. Responsibility of waste generators and local bodies
- Promotion of composting and recycling.

⑦ Briefly explain the following standards and their relevance to a sustainable tech industry:

→ (i) ISO 14040 (Life cycle Assessment):

- Evaluates environmental impact of a product throughout its lifecycle.

(ii) ISO 14020 (Environmental Labeling):

Provides eco-labels to inform consumers about environmentally friendly products

(iii) ISO 14031 (Environmental performance Evaluation):

Helps organizations measure and improve environmental performance